

# Lecture 13: The Dimension

Instructors:

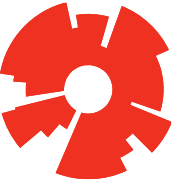
Christoph Lohrmann ([christophl@ru.is](mailto:christophl@ru.is))

*[Based on slides for „Decoding the Matrix“]*



WEEK

SEVEN



# Agenda

## Chapter 6

### Lecture 13:

- Size of a Basis
- Morphing Lemma
- Dimension
- Row Rank and Column Rank
- Subset-Basis Lemma
- Grow Algorithm Termination
- Superset-Basis Lemma
- Dimension Lemma
- Rank Theorem





# Recap: Some Key Take-aways (Week 6)

The **superfluous-vector lemma** states that if a vector in a set of vectors can be represented by a linear combination of the other vectors, then its removal does not affect the span

The **linear-dependence lemma** states that a vector  $v_i$  is in the span of other vectors [and, thus, superfluous for the span] iff the zero vector can be written as a linear combination of the other vectors with  $v_i$ 's coefficient being nonzero

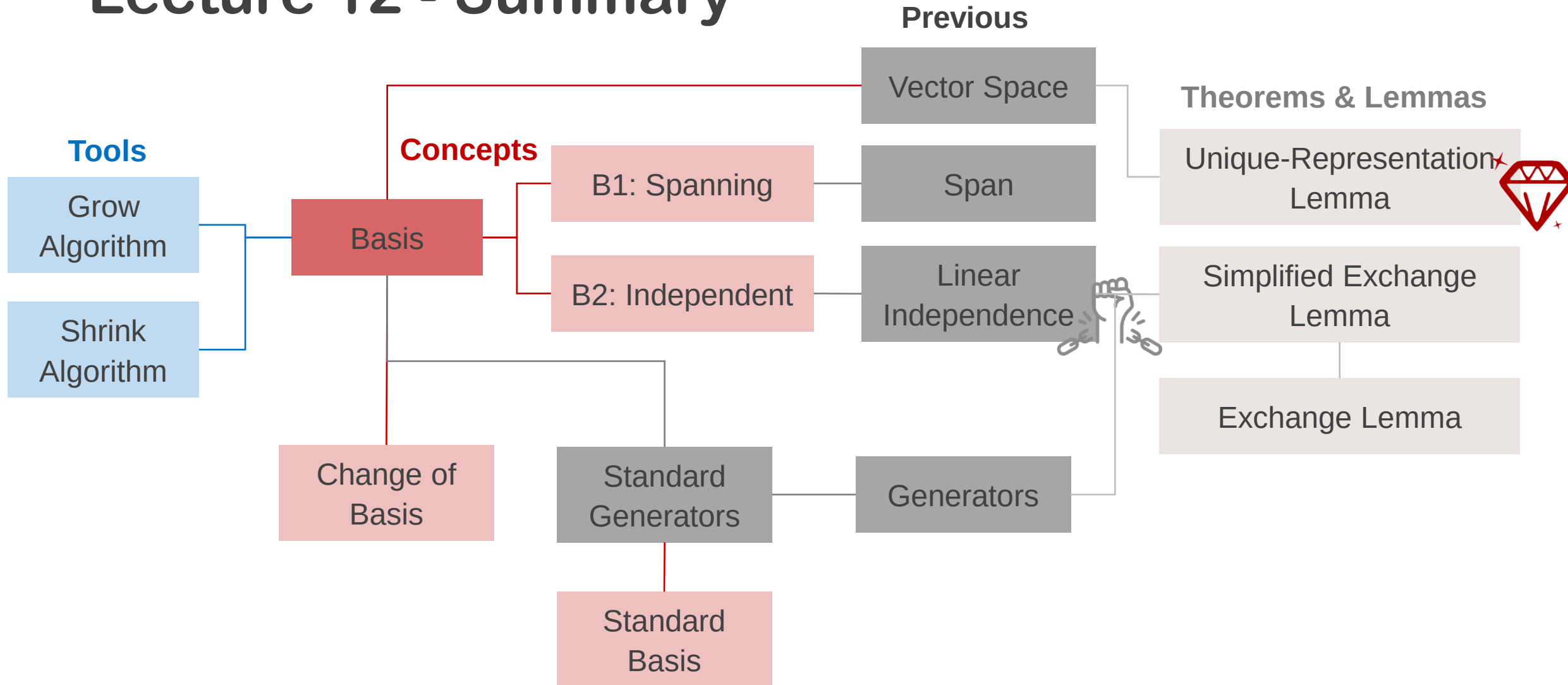
A **basis** for a vector space is a linearly independent set of generators for that vector space. A set of vectors is a basis for a vector space if it satisfies **two properties** – they **span the vector space (B1)** are **linearly independent (B2)**

The **Unique-Representation Lemma** states that for a basis for a vector space there is only one representation of each vector in that vector space using the basis vectors

The **Simplified Exchange Lemma** states that for a set of vectors a vector in that set that is in the span of the other vectors can be removed and another one added, without changing the span. The **Exchange Lemma** additionally adds the requirement that vectors retained and added are linearly independent



# Lecture 12 - Summary





# Size of a Basis

Remember (last lecture)

**Definition:** Let  $\mathcal{V}$  be a vector space. A *basis* for  $\mathcal{V}$  is a linearly independent set of generators for  $\mathcal{V}$ .

Thus a set  $S$  of vectors of  $\mathcal{V}$  is a *basis* for  $\mathcal{V}$  if  $S$  satisfies two properties:

Property B1 (*Spanning*)  $\text{Span } S = \mathcal{V}$ , and

Property B2 (*Independent*)  $S$  is linearly independent.



Key fact for this week: all bases for a vector space have the same size.



We use this as the “basis” for answering many pending questions.



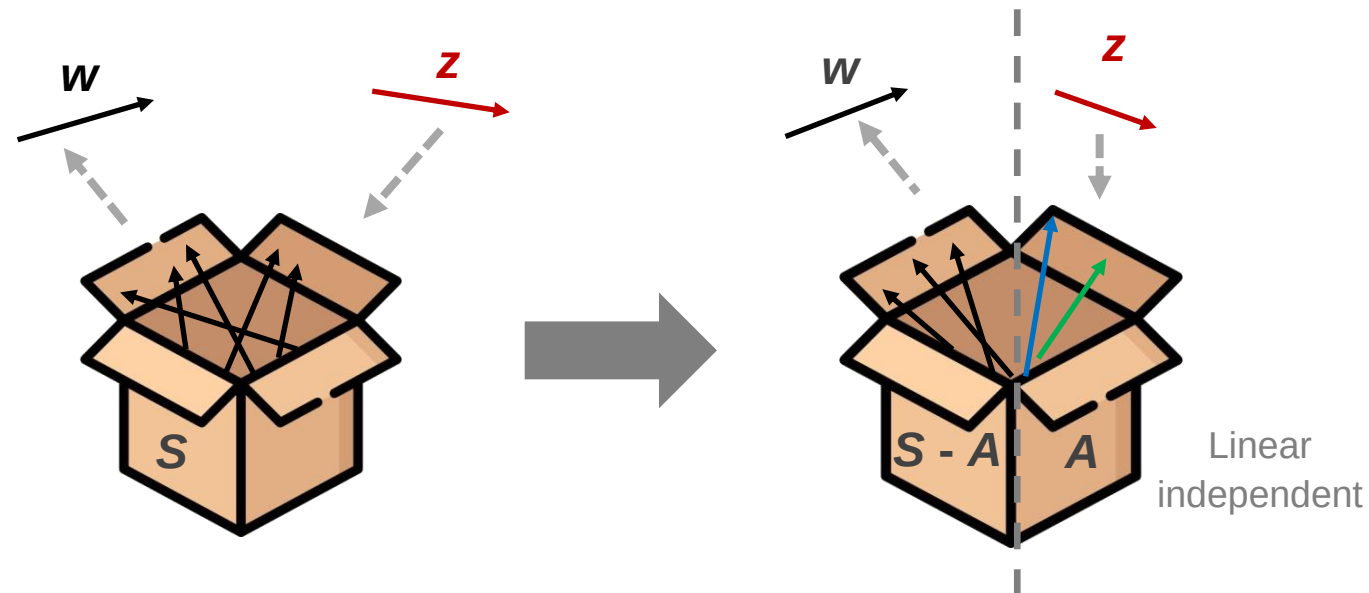
# Morphing Lemma (I)

**Exchange Lemma:** Suppose  $S$  is a set of vectors and  $A$  is a subset of  $S$ . Suppose  $\mathbf{z}$  is a vector in  $\text{Span } S$  such that  $A \cup \{\mathbf{z}\}$  is linearly independent. Then there is a vector  $\mathbf{w} \in S - A$  such that

$$\text{Span } S = \text{Span } (S \overset{\text{in}}{\cup} \{\mathbf{z}\} - \{\mathbf{w}\} \overset{\text{out}}{\})$$

Remember  
(last lecture)

Simplified  
Exchange Lemma



Exchange  
Lemma



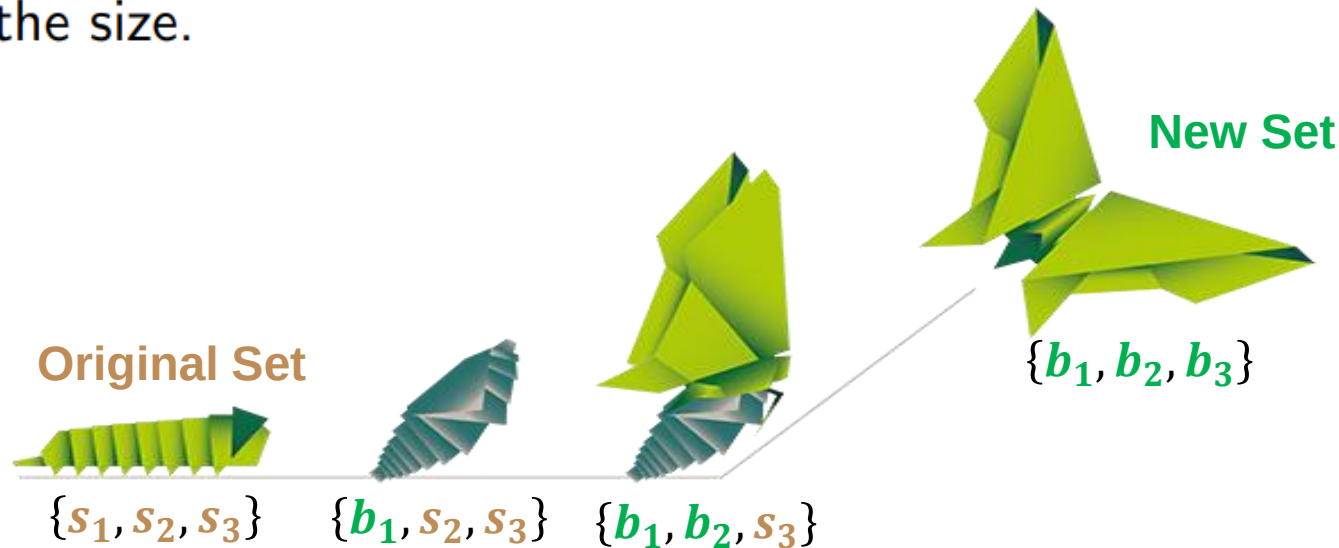


# Morphing Lemma (II)

To morph = "undergo a gradual process of change / transformation"

**Morphing Lemma:** Suppose  $S$  is a set of vectors, and  $B$  is a linearly independent set of vectors in  $\text{Span } S$ . Then  $|S| \geq |B|$ .

Proof outline: modify  $S$  step by step, introducing vectors of  $B$  one by one, without increasing the size.



Span remains the same!  
(Exchange Lemma)

If you take one thing out and put one thing in, then **number** of things remains the same.





# Morphing Lemma (III)



Let  $B = \{\mathbf{b}_1, \dots, \mathbf{b}_n\}$ . Define  $S_0 = S$ .

Prove by induction on  $k \leq n$  that there is a generating set  $S_k$  of  $\text{Span } S$  that contains  $\mathbf{b}_1, \dots, \mathbf{b}_k$  and has size  $|S|$ .

Base case:  $k = 0$  is trivial. ( $S_0 = S$  and  $S$  obviously has  $\text{Span } S$ )

To go from  $S_{k-1}$  to  $S_k$ : use the Exchange Lemma.

►  $A_k = \{\mathbf{b}_1, \dots, \mathbf{b}_{k-1}\}$  and  $\mathbf{z} = \mathbf{b}_k$

Exchange Lemma  $\Rightarrow$  there is a vector  $\mathbf{w}$  in  $S_{k-1}$  such that

$$\underbrace{\text{Span } (S_{k-1} \cup \{\mathbf{b}_k\} - \{\mathbf{w}\})}_{\text{Span } S_k} = \text{Span } S_{k-1}$$

Set  $S_k = S_{k-1} \cup \{\mathbf{b}_k\} - \{\mathbf{w}\}$ . QED

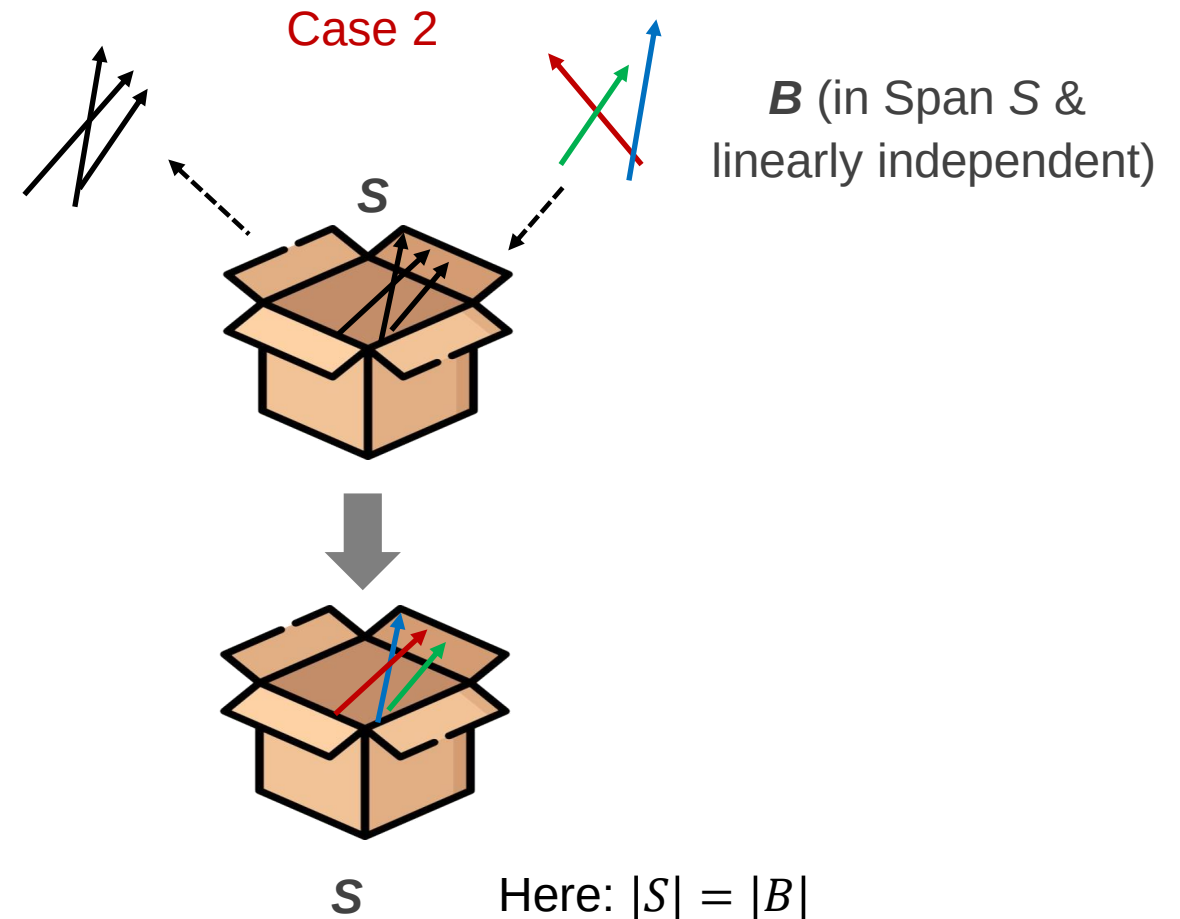
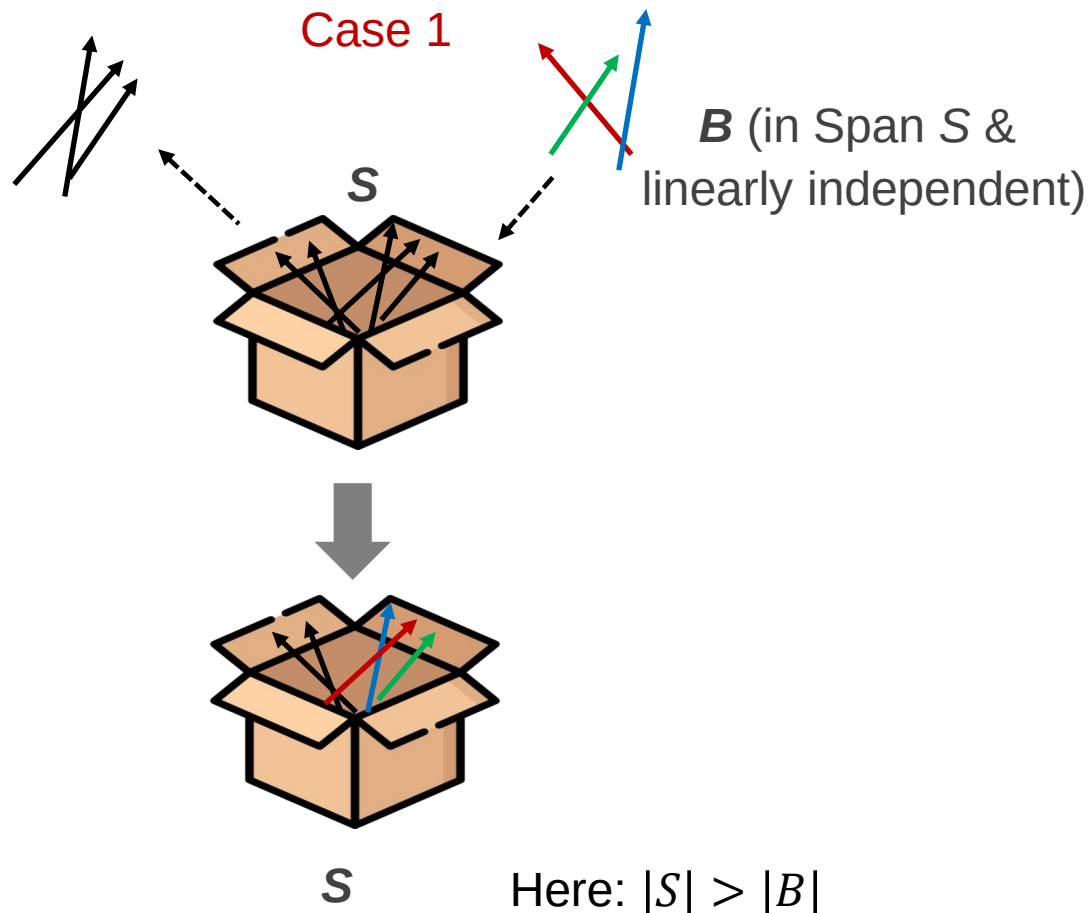
This induction proof is an algorithm.

As we replace vectors in the set until all vectors in set  $B$  of size  $|B|$  are in it, the size of  $S$ ,  $|S|$ , remains the same  $\rightarrow |S| \geq |B|$



# Morphing Lemma (IV)

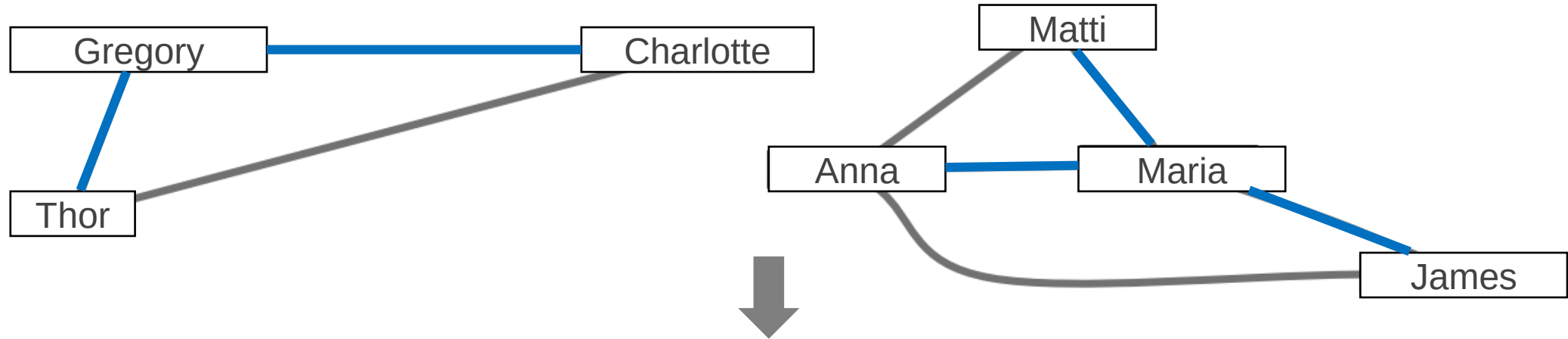
We replace vectors until all vectors in set  $B$  of size  $|B|$  are in it, the size of  $S$ ,  $|S|$ , remains the same  $\rightarrow |S| \geq |B|$



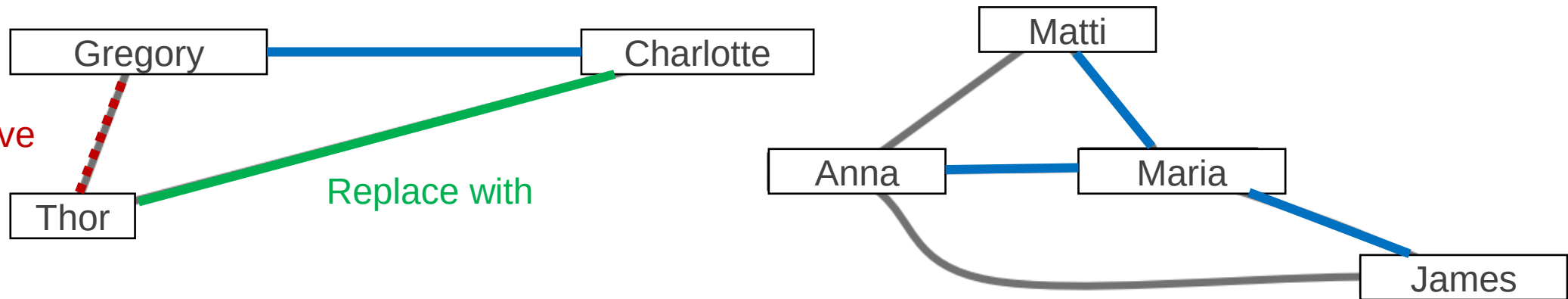


# Morphing from an SF to another SF (I)

Starting SF (is Spanning)



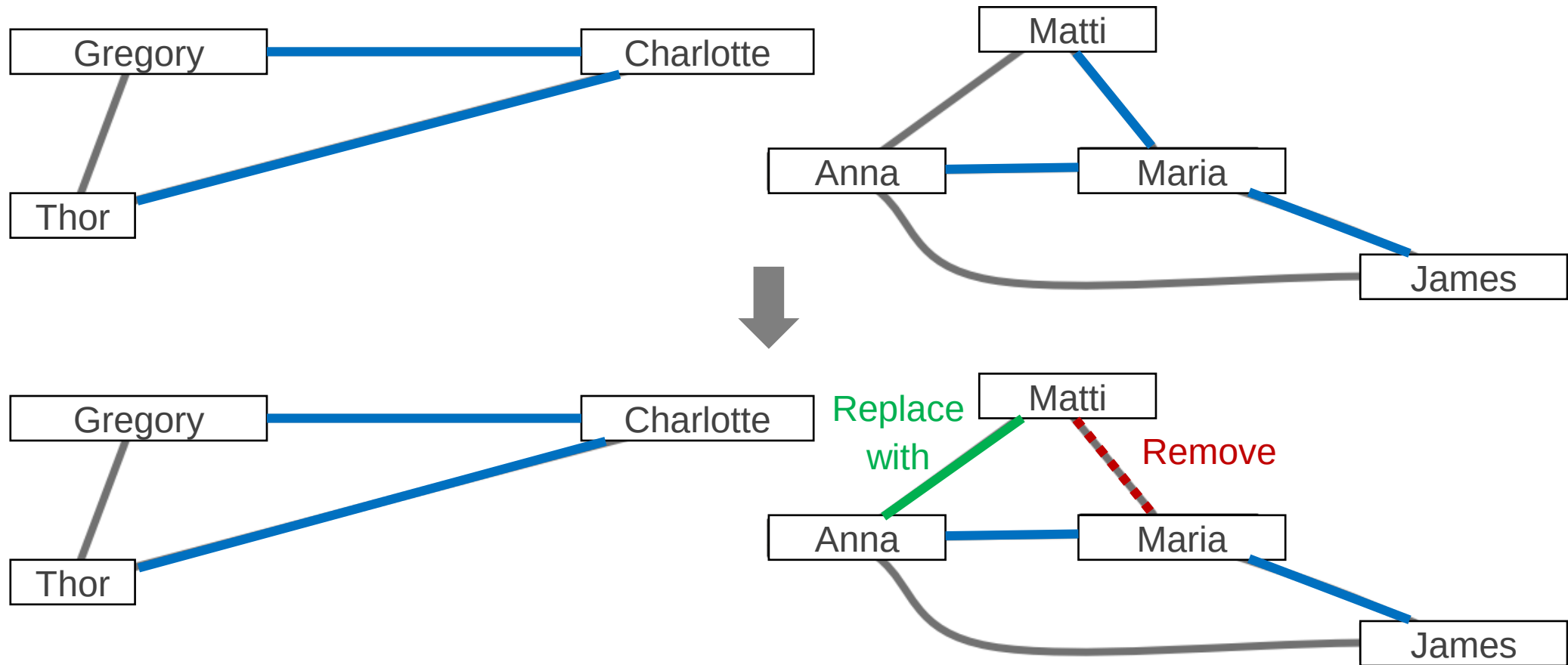
Remove



We can still reach everyone → still Spanning



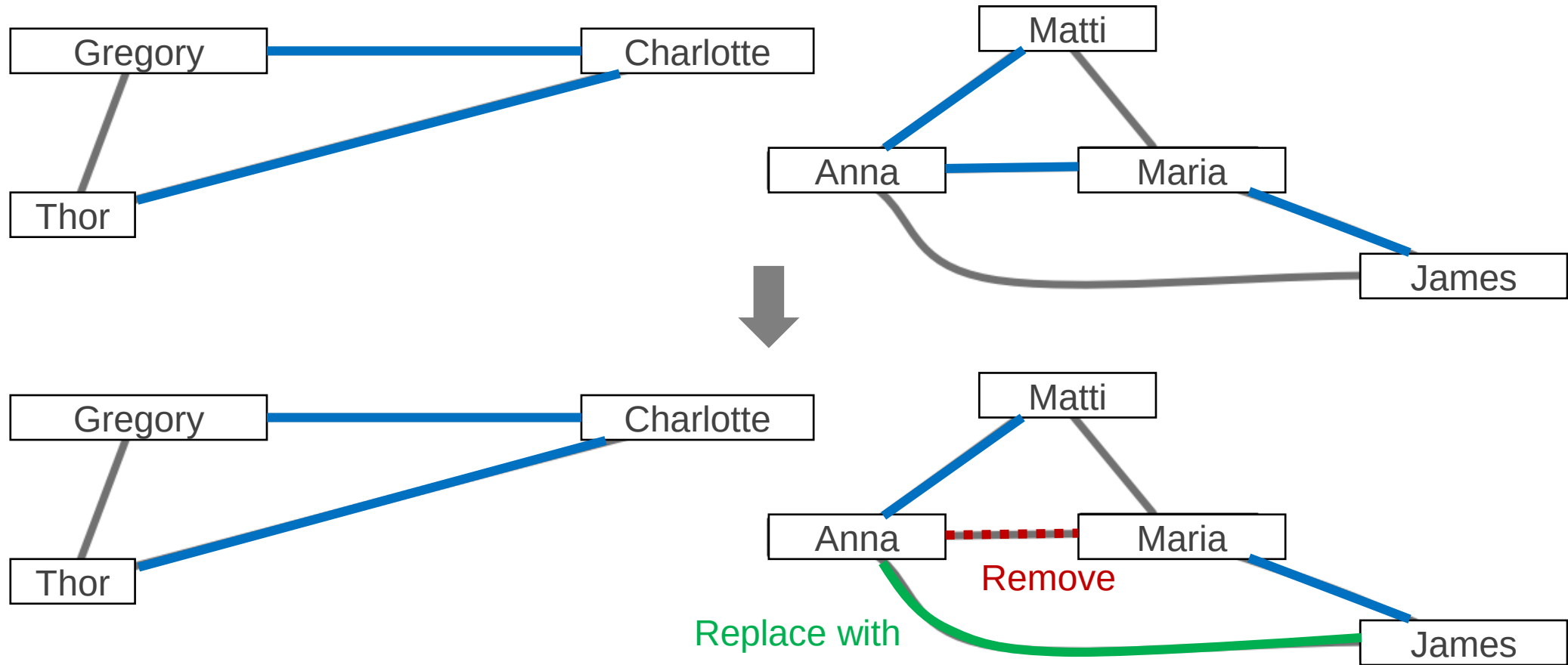
# Morphing from an SF to another SF (II)



We can still reach everyone → still Spanning



# Morphing from an SF to another SF (III)

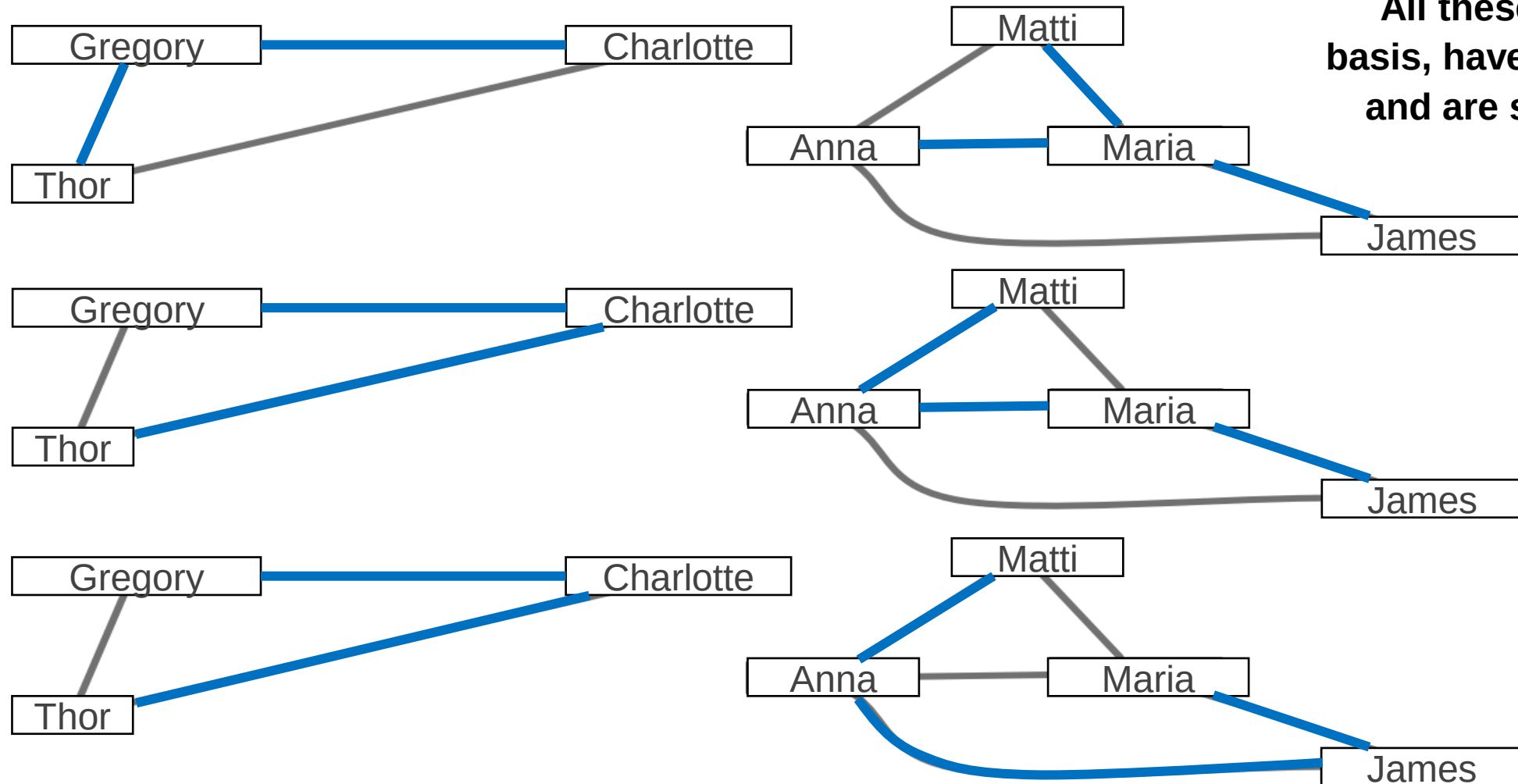


We can still reach everyone → still Spanning



# Morphing from an SF to another SF (IV)

**All these SF are basis, have same size and are spanning**





# Morphing Lemma - Generating Set Size

**Morphing Lemma:** Suppose  $S$  is a set of vectors, and  $B$  is a linearly independent set of vectors in  $\text{Span } S$ . Then  $|S| \geq |B|$ .



**Theorem:** Any basis for  $\mathcal{V}$  is a smallest generating set for  $\mathcal{V}$ .



**Proof:** Let  $S$  be a smallest generating set for  $\mathcal{V}$ . Let  $B$  be a basis for  $\mathcal{V}$ . Then  $B$  is a linearly independent set of vectors in  $\text{Span } S$ . By the **Morphing Lemma**,  $B$  is no bigger than  $S$ , so  $B$  is also a smallest generating set.

**Theorem:** All bases for a vector space  $\mathcal{V}$  have the same size.



**Proof:** They are all smallest generating sets.  
(See also previous proof)



# Dimension

**Definition:** We define the dimension of a vector space to be the size of a basis for that vector space. The dimension of a vector space  $\mathcal{V}$  is written  $\dim \mathcal{V}$ .

**Definition:** We define the rank of a set  $S$  of vectors as the dimension of  $\text{Span } S$ . We write  $\text{rank } S$ .

(A set of vectors is not a vector space, but the span of them is a vector space)

---

**Example:** The vectors  $[1, 0, 0]$ ,  $[0, 2, 0]$ ,  $[2, 4, 0]$  are linearly dependent.

Therefore their rank is less than three.

First two of these vectors form a basis for the span of all three, so the rank is two.

**Example:** The vector space  $\text{Span } \{[0, 0, 0]\}$  is spanned by an empty set of vectors.

Therefore the rank of  $\{[0, 0, 0]\}$  is zero.



# Row Rank and Column Rank (I)

**Definition:** For a matrix  $M$ , the row rank of  $M$  is the rank of its rows, and the column rank of  $M$  is the rank of its columns.

Equivalently, the row rank of  $M$  is the dimension of Row  $M$ , and the column rank of  $M$  is the dimension of Col  $M$ .

**Example:** Consider the matrix

$$M = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 2 & 4 & 0 \end{bmatrix}$$

← In the span of the first two rows, thus, **row rank is two**

$$M = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 2 & 4 & 0 \end{bmatrix}$$

Zero vector does not span the column space, thus, **column rank is two**



# Row Rank and Column Rank (II)

**Example:** Consider the matrix [Not a square matrix]

$$M = \begin{bmatrix} 1 & 0 & 0 & 5 \\ 0 & 2 & 0 & 7 \\ 0 & 0 & 3 & 9 \end{bmatrix}$$

All three rows are linearly independent (none is in the span of the others), thus, **row rank is three**

$$M = \begin{bmatrix} 1 & 0 & 0 & 5 \\ 0 & 2 & 0 & 7 \\ 0 & 0 & 3 & 9 \end{bmatrix}$$

The first three columns are linearly independent, but ...

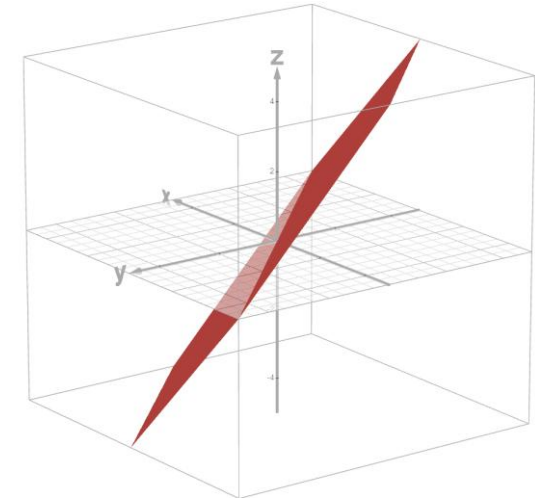
... this vector can be written as a linear combination of the first three vectors, thus, **column rank is (just) three**

Does column rank always equal row rank?



# Row Rank and Column Rank (III)

For any set of vectors  $S$ , the rank  $S \leq |S|$



## Examples:

- Span  $\{[1, 2, -2]\}$  is a line but Span  $\{[0, 0, 0]\}$  is a point.  
First vector space has dimension one, second has dimension zero.

One vector, but  
dimension **one** or **less**

- Span  $\{[1, 2], [3, 4]\}$  consists of all of  $\mathbb{R}^2$  but Span  $\{[1, 3], [2, 6]\}$  is a line  
The first has dimension two and the second has dimension one.

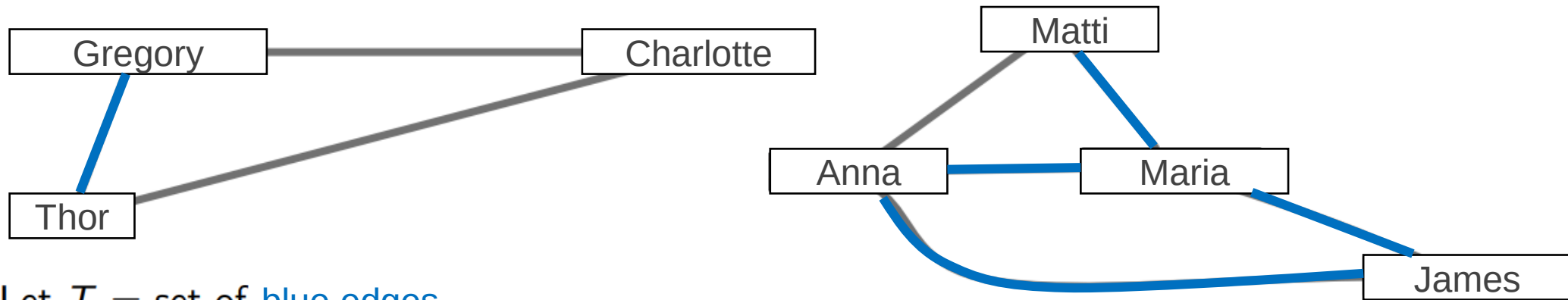
Two vectors, but  
dimension **two** or **less**

- Span  $\{[1, 0, 0], [0, 1, 0], [0, 0, 1]\}$  is  $\mathbb{R}^3$  but Span  $\{[1, 0, 0], [0, 1, 0], [1, 1, 0]\}$  is a plane.  
The first has dimension three and the second has dimension two.

Three vectors, but  
dimension **three** or **less**

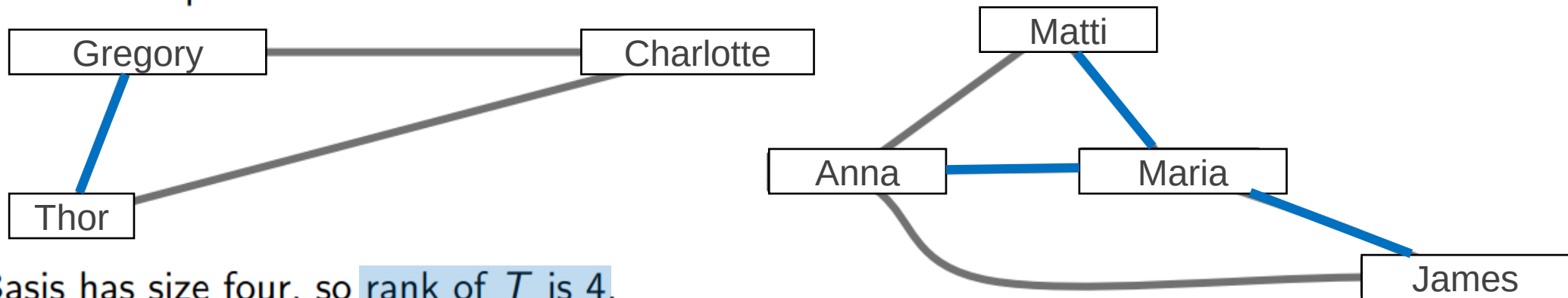


# Dimension and Rank in Graphs (I)



Let  $T$  = set of blue edges

Basis for Span  $T$ :

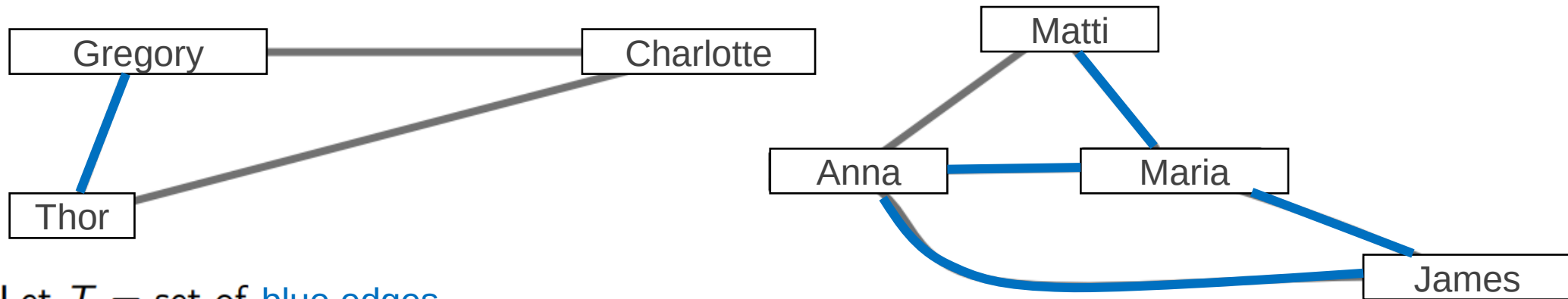


Basis has size four, so rank of  $T$  is 4.

[Note: in this example, we do not „reach“ Charlotte]

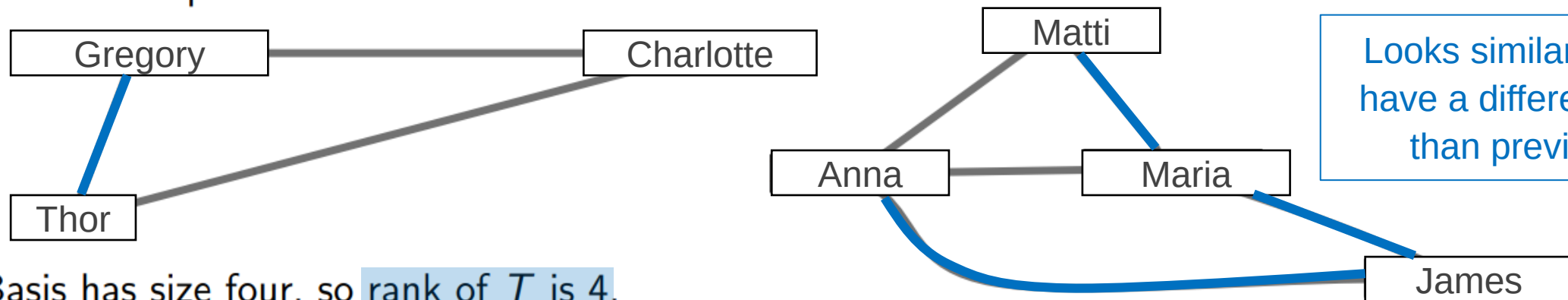


# Dimension and Rank in Graphs (II)



Let  $T$  = set of blue edges

Basis for Span  $T$ :



Looks similar, but we have a different basis than previously

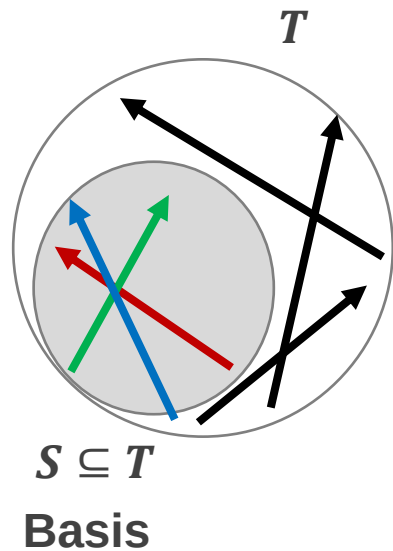
Basis has size four, so rank of  $T$  is 4.

[Note: in this example, we do not „reach“ **Charlotte**]



# Subset-Basis Lemma (I)

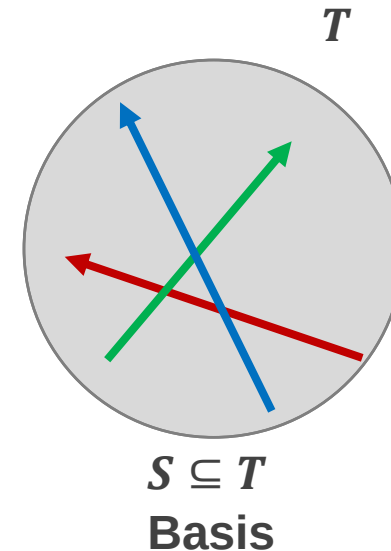
**Lemma:** Every finite set  $T$  of vectors contains a subset  $S$  that is a basis for  $\text{Span } T$ .



Set of Vectors  $T$

$[2, 0, 0]$   
 $[0, 1, 0]$   
 $[0, 1, 2]$   
 $[1, 3, 0]$   
 $[2, 2, 2]$

**Basis for**  
 $\text{Span } T$



Set of Vectors  $T$

$[2, 1, 0]$   
 $[0, 1, 0]$   
 $[0, 2, 1]$

**Basis for**  
 $\text{Span } T$



# Subset-Basis Lemma (II)

**Lemma:** Every finite set  $T$  of vectors contains a subset  $S$  that is a basis for  $\text{Span } T$ .



**Proof:** The Grow algorithm finds a basis for  $\mathcal{V}$  if it terminates.



Initialize  $S = \emptyset$ .

Repeat while possible: select a vector  $\mathbf{v}$  in  $\mathcal{V}$  that is not in  $\text{Span } S$ , and put it in  $S$ .



Revised version:

(using the vectors in our set)

Initialize  $S = \emptyset$

Repeat while possible: select a vector  $\mathbf{v}$  in  $T$  that is not in  $\text{Span } S$ , and put it in  $S$ .





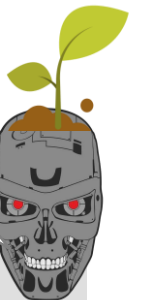
# Grow Algorithm Termination (I)

```
def GROW( $\mathcal{V}$ )
   $B = \emptyset$ 
  repeat while possible:
    find a vector  $\mathbf{v}$  in  $\mathcal{V}$  that is not in  $\text{Span } B$ , and put it in  $B$ .
```



I add vectors to grow the Span until I terminate

**Grow-Algorithm-Termination Lemma:** If  $\mathcal{V}$  is a subspace of  $\mathbb{F}^D$  where  $D$  is finite then  $\text{GROW}(\mathcal{V})$  terminates.



**Proof:** By Grow-Algorithm Corollary,  $B$  is linearly independent throughout.

Apply the Morphing Lemma with  $S = \{\text{standard generators for } \mathbb{F}^D\} \Rightarrow$   
 $|B| \leq |S| = |D|.$  (Simple, extreme Case)

Since  $B$  grows in each iteration, there are at most  $|D|$  iterations.

QED



# Grow Algorithm Termination (II)

```
def GROW( $\mathcal{V}$ )
   $B = \emptyset$ 
  repeat while possible:
    find a vector  $\mathbf{v}$  in  $\mathcal{V}$  that is not in  $\text{Span } B$ , and put it in  $B$ .
```



(After termination, set  $B$  is a Basis for  $\mathcal{V}$ )

**Theorem:** For finite  $D$ , every subspace of  $\mathbb{F}^D$  contains a basis.



Grow-Algorithm-Termination Lemma ensures algorithm terminates.

(B1: Spanning)

Upon termination, every vector in  $\mathcal{V}$  is in  $\text{Span } B$ , so  $B$  is a set of generators for  $\mathcal{V}$ . By Grow-Algorithm Corollary,  $B$  is linearly independent. Therefore  $B$  is a basis for  $\mathcal{V}$ . QED

(B2: Linear Independence)

(B1+B2)



# Grow Algorithm Termination (III)

Assume we have a vector space  $\mathcal{V}$  that is spanned by three vectors, which is a subspace of  $\mathbb{R}^4$ . The three vectors are  $\{[2, 0, 0, 0], [0, 1, 0, 0], [1, 3, 0, 0]\}$ .

Obviously, since we have 3 vectors, there rank is at most 3 (three linearly independent vectors)

Since our vector space  $\mathcal{V}$  is a subspace of  $\mathbb{R}^4$ , we know that there are at most four additions (= adding vector not in span) of the **Grow algorithm**.

Let's run the Grow algorithm (with the vectors we have)



# Grow Algorithm Termination (IV)

Assume we have a vector space  $\mathcal{V}$  that is spanned by three vectors, which is a subspace of  $\mathbb{R}^4$ . The three vectors are  $\{[2, 0, 0, 0], [0, 1, 0, 0], [1, 3, 0, 0]\}$ .

## Grow algorithm:

- (1) We add  $[2, 0, 0, 0]$  as it is not in the span of the empty set,
  - (2) then we add  $[0, 1, 0, 0]$  as it is not within the  $\text{Span}([2, 0, 0, 0])$
  - (3) We do NOT add  $[1, 3, 0, 0]$  as it is within the  $\text{Span}([2, 0, 0, 0], [0, 1, 0, 0])$
- (No more vectors)



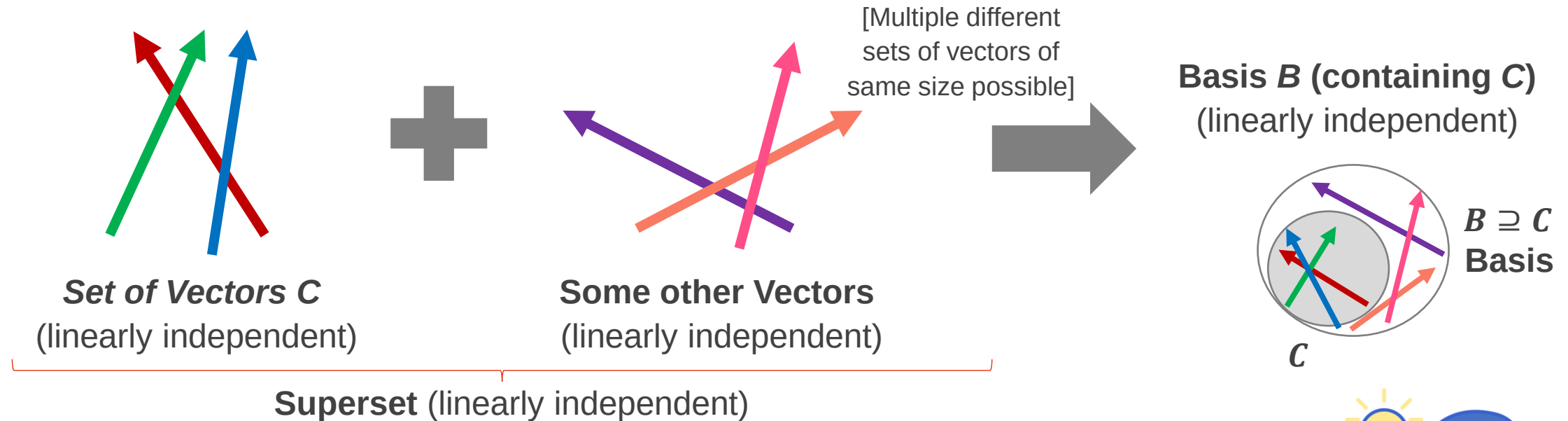
I terminated!

Every Vector in  $\mathcal{V}$  is in  $\text{Span}([2, 0, 0, 0], [0, 1, 0, 0])$  **[B1]** and  
these vectors are linearly independent **[B2]**  
→  $\text{Span}([2, 0, 0, 0], [0, 1, 0, 0])$  is a **basis** for  $\mathcal{V}$



# Not Subset Superset-Basis Lemma (I)

**Superset-Basis Lemma:** Let  $\mathcal{V}$  be a vector space consisting of  $D$ -vectors where  $D$  is finite. Let  $C$  be a linearly independent set of vectors belonging to  $\mathcal{V}$ . Then  $\mathcal{V}$  has a basis  $B$  containing all vectors in  $C$ .



So, if we have a set of linearly independent vectors, there is at least one if not multiple basis that we can build with them and some additional vectors.

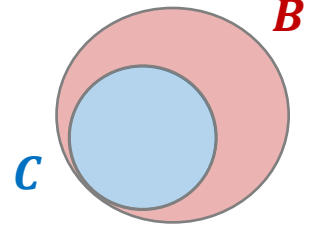




# Not Subset

## Superset-Basis Lemma (II)

**Superset-Basis Lemma:** Let  $\mathcal{V}$  be a vector space consisting of  $D$ -vectors where  $D$  is finite. Let  $C$  be a linearly independent set of vectors belonging to  $\mathcal{V}$ . Then  $\mathcal{V}$  has a basis  $B$  containing all vectors in  $C$ .



**Proof:** Use version of Grow algorithm:

Initialize  $B$  to the empty set.

Repeat while possible: select a vector  $\mathbf{v}$  in  $\mathcal{V}$  (preferably in  $C$ ) that is not in  $\text{Span } B$ , and put it in  $B$ .

At first,  $B$  will consist of vectors in  $C$  until  $B$  contains all of  $C$ . Then more vectors will be added to  $B$  until  $\text{Span } B = \mathcal{V}$ . By Grow-Algorithm Corollary,  $B$  is linearly independent throughout. Therefore, once algorithm terminates,  $B$  contains  $C$  and is a basis for  $\mathcal{V}$ .

**Superset**      **Subset**

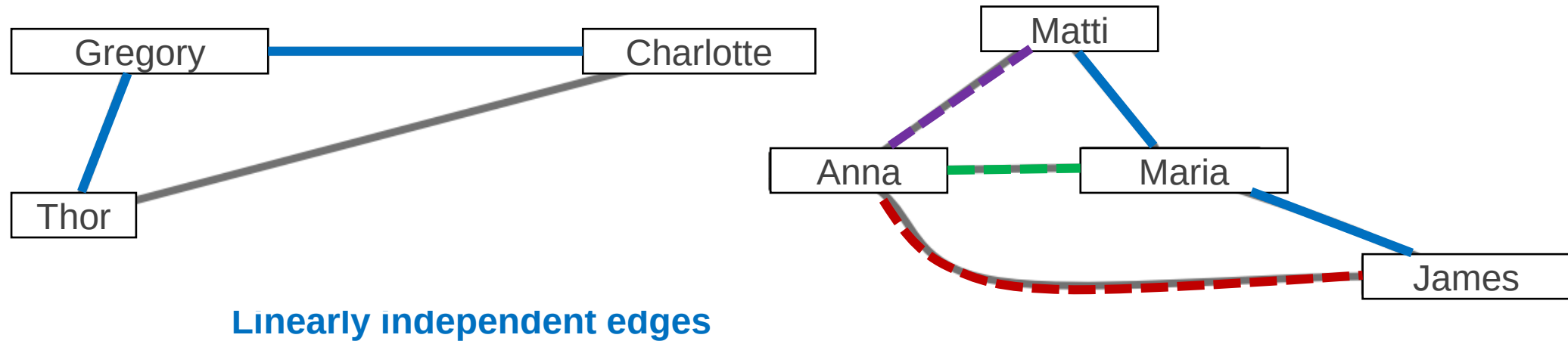
Reach  $B = C$   
and then  
continue until  
 $\text{Span } B = \mathcal{V}$

Termination is implied by Grow Algorithm Termination Lemma.

**QED**



# Not Subset Superset-Basis Lemma (III)



We can add a different additional edge to Span

— — — or — — — or — — —

Not just B2 (Linear Independence) but then  
also B1 (Spanning) → **Basis**

So there is at least one (here: three)  
**superset** of edges that forms a basis  
with our **blue edges** (linearly  
independent) as a subset



# Not Subset Superset-Basis Lemma (IV)

Assume we have a vector space  $\mathcal{V}$  that is  $\mathbb{R}^4$ . And we have the following three linearly independent vectors  $\{[2, 0, 0, 0], [0, 1, 0, 0], [0, 3, 1, 0]\}$ .

Based on the **Superset-Basis Lemma**, there is a basis for  $\mathbb{R}^4$  that contains these linearly independent vectors.

A simple example would be

$\{[2, 0, 0, 0], [0, 1, 0, 0], [0, 3, 1, 0], [0, 0, 0, 1]\}$  or

.... or

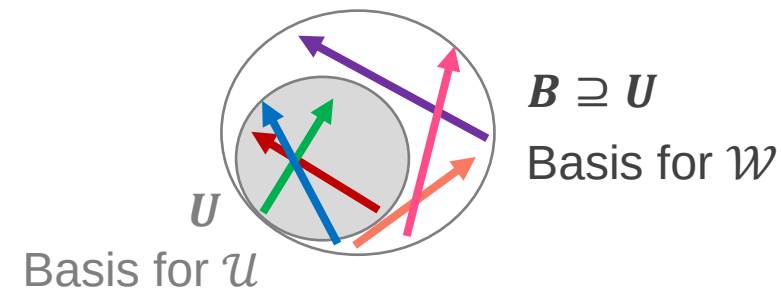
$\{[2, 0, 0, 0], [0, 1, 0, 0], [0, 3, 1, 0], [4, 0, 0, 5]\}$

as these all span  $\mathbb{R}^4$ , our vector space [B1], and are linearly independent [B2], they are a **basis** for  $\mathbb{R}^4$

(You can use the vectors to create the standard generators for  $\mathbb{R}^4$ , so they obviously span  $\mathbb{R}^4$ )



# Dimension Lemma (I)



If  $\mathcal{U}$  and  $\mathcal{W}$  are vector spaces and  $\mathcal{U}$  is a subset of  $\mathcal{W}$ , we say  $\mathcal{U}$  is a **subspace** of  $\mathcal{W}$

**Dimension Lemma:** If  $\mathcal{U}$  is a subspace of  $\mathcal{W}$  then

- ▶ **D1:**  $\dim \mathcal{U} \leq \dim \mathcal{W}$ , and
- ▶ **D2:** if  $\dim \mathcal{U} = \dim \mathcal{W}$  then  $\mathcal{U} = \mathcal{W}$

If  $\mathcal{U}$  is a subspace of  $\mathcal{W}$  then the dimension of  $\mathcal{U}$  can not be larger than the dimension of  $\mathcal{W}$



**Proof:** Let  $\mathbf{u}_1, \dots, \mathbf{u}_k$  be a basis for  $\mathcal{U}$ .

By Superset-Basis Lemma, there is a basis  $B$  for  $\mathcal{W}$  that contains  $\mathbf{u}_1, \dots, \mathbf{u}_k$ .

- ▶  $B = \{\mathbf{u}_1, \dots, \mathbf{u}_k, \mathbf{b}_1, \dots, \mathbf{b}_r\}$
- ▶ Thus  $k \leq |B|$ , and
- ▶ If  $k = |B|$  then  $\{\mathbf{u}_1, \dots, \mathbf{u}_k\} = B$

**QED**



# Dimension Lemma (II)

**Proposition:** Any set of  $D$ -vectors has rank at most  $|D|$ .

**Example:**  $S = \{[-0.6, -2.1, -3.5, -2.2], [-1.3, 1.5, -0.9, -0.5], [4.9, -3.7, 0.5, -0.3], [2.6, -3.5, -1.2, -2.0], [-1.5, -2.5, -3.5, 0.94]\}$  **4-vectors**

Since every vector in  $S$  is a 4-vector,  $\text{Span } S$  is a subspace of  $\mathbb{R}^4$ .

Since  $\dim \mathbb{R}^4 = 4$ , D1 shows  $\dim \text{Span } S \leq 4$ .



Would be nice to be more precise ...



# Rank Theorem

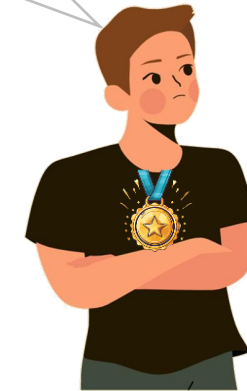
Should I calculate the row rank or the column rank?

Don't worry Gary, it does not matter which one!



**Rank Theorem:** For every matrix  $M$ , row rank equals column rank.

**Lemma:** For any matrix  $A$ , row rank of  $A \leq$  column rank of  $A$   
To show theorem:



**Part 1** ▶ Apply lemma to  $M \Rightarrow$  row rank of  $M \leq$  column rank of  $M$

**Part 2** ▶ Apply lemma to  $M^T \Rightarrow$  row rank of  $M^T \leq$  column rank of  $M^T \Rightarrow$  column rank of  $M \leq$  row rank of  $M$

Combine  $\Rightarrow$  row rank of  $M =$  column rank of  $M$



## Part 1

# Rank Theorem - The Proof (I)

We got to prove it now too?  
That will be exhausting ...

It will be fine Gary, you can  
do it!



Think of  $A$  as columns  $\mathbf{a}_1, \dots, \mathbf{a}_n$ .  
Let  $\mathbf{b}_1, \dots, \mathbf{b}_r$  be basis for column space

Column rank =  $r$

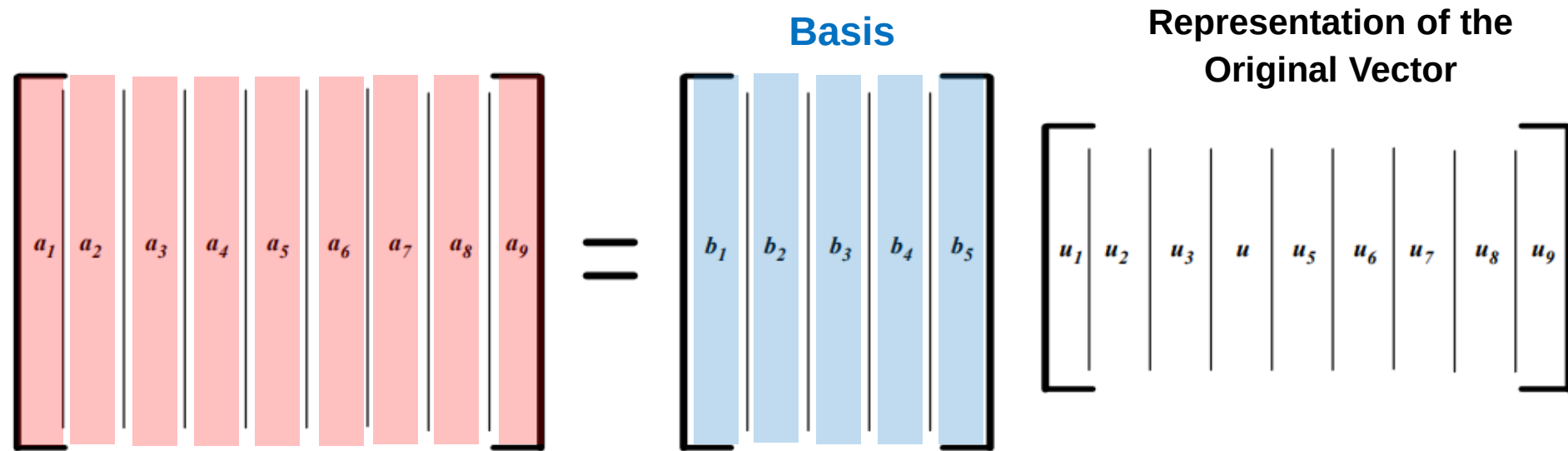




Part 1

# Rank Theorem - The Proof (II)

rewrite as matrix-matrix equation  $A = BU$



$r$  columns

$r$  rows  $\times n$  columns

Needs to match to avoid the math police

Main takeaway:  $U$  has  $r$  rows

I'm watching you!



**Math Police**



## Part 1

# Rank Theorem - The Proof (III)

Let's look at it from the row perspective

Linear combination of  
the  $r$  rows of  $U$

$$\begin{bmatrix} \bar{a}_1 \\ \bar{a}_2 \\ \bar{a}_3 \\ \bar{a}_4 \\ \bar{a}_5 \\ \bar{a}_6 \\ \bar{a}_7 \end{bmatrix} = \begin{bmatrix} \bar{b}_1 \\ \bar{b}_2 \\ \bar{b}_3 \\ \bar{b}_4 \\ \bar{b}_5 \\ \bar{b}_6 \\ \bar{b}_7 \end{bmatrix} \begin{bmatrix} \bar{u}_1 \\ \bar{u}_2 \\ \bar{u}_3 \\ \bar{u}_4 \\ \bar{u}_5 \end{bmatrix}$$

These rows could be linearly independent (row rank is  $r$ ) or they are not (row rank is smaller  $r$ )

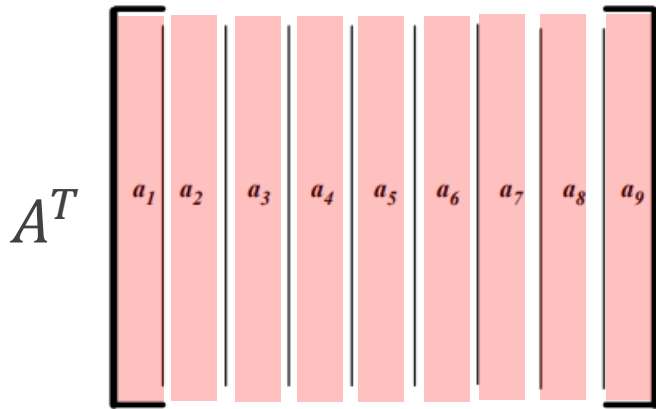
**Thus row rank of  $A$  is at most  $r$  which is column rank of  $A$**

$$\text{row rank of } M \leq \text{column rank of } M$$



Part 2

# Rank Theorem - The Proof (IV)



How about doing now  
the same with  $A^T$ ?

Great idea, Gary!



Let  $\mathbf{b}_1, \dots, \mathbf{b}_r$  be basis for column space of  $A^T$  (so column rank =  $r$ ).  $\rightarrow$  row rank =  $r$  of  $A$

It's a transpose, so these two are the same



## Part 2

## Rank Theorem - The Proof (V)

Let's look at it from the row perspective

$$A^T \begin{bmatrix} \bar{a}_1 \\ \bar{a}_2 \\ \bar{a}_3 \\ \bar{a}_4 \\ \bar{a}_5 \\ \bar{a}_6 \\ \bar{a}_7 \end{bmatrix} = \begin{bmatrix} \bar{b}_1 \\ \bar{b}_2 \\ \bar{b}_3 \\ \bar{b}_4 \\ \bar{b}_5 \\ \bar{b}_6 \\ \bar{b}_7 \end{bmatrix} \begin{bmatrix} \bar{u}_1 \\ \bar{u}_2 \\ \bar{u}_3 \\ \bar{u}_4 \\ \bar{u}_5 \end{bmatrix}$$

Each row of  $A^T$  is in span of the  $r$  rows of  $U$ . Thus row rank of  $A^T$  is at most  $r$  which is column rank of  $A^T$

→ Thus, the column rank of  $A$  is at most  $r$ , the row rank of  $A$

$$\text{column rank of } M \leq \text{row rank of } M$$





# Rank Theorem - The Proof (VI)

I'm at least as grumpy  
as you are!

$$\text{Grumpy}_1 \geq \text{Grumpy}_2$$

And I'm at least as  
grumpy as you are!

$$\text{Grumpy}_2 \geq \text{Grumpy}_1$$

I guess that means we  
are equally grumpy!

$$\text{Grumpy}_2 = \text{Grumpy}_1$$

Whatever





# Rank Theorem - The Proof (VII)

**Part 1** row rank of  $A$  is at most  $r$  the column rank of  $A$

Row rank  $A \leq$  Column rank  $A$

**Part 2** column rank of  $A$  is at most  $r$ , the row rank of  $A$

Column rank  $A \leq$  Row rank  $A$

Yep, Gary. But hey, that means one less thing to bother about when determining the rank



All of this, so we just need to determine one of them e.g., the column rank?



**Column rank  $A =$  Row rank  $A$**



# Exercise: Rank & Dimension

You are given the following vectors:

- 1) Assume that a matrix  $A$  has  $a_1, a_2, a_3$  as column vectors. What is the column rank of the vectors in  $A$ ? Do they form a basis for  $\mathbb{R}^3$ ? Show work.
- 2) Show using a calculation that for matrix  $A$  the row rank equals the column rank (from (1)).
- 3) Without doing any calculation, what is the column rank of  $A^T$ ? Explain.
- 4) Assume that a matrix  $B$  has  $b_1, b_2, b_3$  as column vectors and that they form the basis for a 3-dimensional subspace in  $\mathbb{R}^4$ . Will adding  $b_4$  to matrix  $B$  change that the column vectors in  $B$  is a basis for the original 3D subspace?

$$a_1 = \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix} \quad a_2 = \begin{bmatrix} 0.5 \\ 3 \\ 1 \end{bmatrix} \quad a_3 = \begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix}$$

$$b_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} \quad b_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix} \quad b_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix} \quad b_4 = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix}$$



## Solution

## Exercise: Rank &amp; Dimension

$$a_1 = \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix} \quad a_2 = \begin{bmatrix} 0.5 \\ 3 \\ 1 \end{bmatrix} \quad a_3 = \begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix}$$

- a) We could use the Grow or Shrink algorithm to find a basis for the span of these three vectors. Using the starting point of all vectors, we can first determine, if first, we they are linearly independent.

To determine whether  $a_1$ ,  $a_2$ , and  $a_3$  are **linearly independent**, we need to show that the zero vector can **only** be written via a trivial linear combination (= all alpha coefficients zero) of these vectors. So,

$$\mathbf{0} = \alpha_1 a_1 + \alpha_2 a_2 + \alpha_3 a_3 \quad \text{only with} \quad \alpha_1, \alpha_2, \alpha_3 = 0$$

$$[0, 0, 0] = \alpha_1 [2, 1, 0] + \alpha_2 [0.5, 3, 1] + \alpha_3 [0, 1, 2]$$

$$= [2\alpha_1 + 0.5\alpha_2, \alpha_1 + 3\alpha_2 + \alpha_3, \alpha_2 + 2\alpha_3]$$

Set of linear equations to solve

$$\begin{array}{c}
 2\alpha_1 + 0.5\alpha_2 = 0 \\
 \alpha_1 + 3\alpha_2 + \alpha_3 = 0 \\
 \alpha_2 + 2\alpha_3 = 0
 \end{array}
 \begin{array}{c}
 \longleftrightarrow \\
 R_2 - 0.5R_3
 \end{array}
 \begin{array}{c}
 2\alpha_1 + 0.5\alpha_2 = 0 \\
 \alpha_1 + 2.5\alpha_2 = 0 \\
 \alpha_2 + 2\alpha_3 = 0
 \end{array}
 \begin{array}{c}
 \longleftrightarrow \\
 R_2 - 5R_1
 \end{array}
 \begin{array}{c}
 2\alpha_1 + 0.5\alpha_2 = 0 \\
 -9\alpha_1 = 0 \\
 \alpha_2 + 2\alpha_3 = 0
 \end{array}
 \begin{array}{c}
 \longleftrightarrow \\
 2(0) + 0.5\alpha_2 = 0
 \end{array}
 \begin{array}{c}
 \longleftrightarrow \\
 \alpha_2 = 0 \\
 \alpha_1 = 0 \\
 0 + 2\alpha_3 = 0
 \end{array}
 \begin{array}{c}
 \longleftrightarrow \\
 \alpha_1 = 0 \\
 \alpha_2 = 0 \\
 \alpha_3 = 0
 \end{array}$$



Solution

## Exercise: Rank & Dimension

$$a_1 = \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix} \quad a_2 = \begin{bmatrix} 0.5 \\ 3 \\ 1 \end{bmatrix} \quad a_3 = \begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix}$$

The only linear combination is the trivial one. We can conclude that  $a_1, a_2, a_3$  are **linearly independent**. Thus, the three linearly independent vectors have a column rank of 3. Since we have three linearly independent 3-vectors, we can conclude that they are a basis for  $\mathbb{R}^3$ .

- b) For this task, the vector change – let's call them  $r_1, r_2$ , and  $r_3$  as these are row vectors. To determine whether  $r_1, r_2$ , and  $r_3$  are **linearly independent**, we need to show that the zero vector can **only** be written via a trivial linear combination (= all alpha coefficients zero) of these vectors. So,

$$\mathbf{0} = \alpha_1 r_1 + \alpha_2 r_2 + \alpha_3 r_3 \quad \text{only with} \quad \alpha_1, \alpha_2, \alpha_3 = 0$$

$$[0, 0, 0] = \alpha_1 [2, 0.5, 0] + \alpha_2 [1, 3, 1] + \alpha_3 [0, 1, 2]$$

$$= [2\alpha_1 + \alpha_2, 0.5\alpha_1 + 3\alpha_2 + \alpha_3, \alpha_2 + 2\alpha_3]$$



Solution

## Exercise: Rank &amp; Dimension

$$a_1 = \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix} \quad a_2 = \begin{bmatrix} 0.5 \\ 3 \\ 1 \end{bmatrix} \quad a_3 = \begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix}$$

$$[0, 0, 0] = [2\alpha_1 + \alpha_2, 0.5\alpha_1 + 3\alpha_2 + \alpha_3, \alpha_2 + 2\alpha_3]$$

Set of linear equations to solve

$$\begin{array}{c} 2\alpha_1 + \alpha_2 = 0 \\ 0.5\alpha_1 + 3\alpha_2 + \alpha_3 = 0 \\ \alpha_2 + 2\alpha_3 = 0 \end{array} \quad \begin{array}{c} \longleftrightarrow \\ \\ \end{array} \quad \begin{array}{c} 2\alpha_1 + 0.5\alpha_2 = 0 \\ 0.5\alpha_1 + 2.5\alpha_2 = 0 \\ \alpha_2 + 2\alpha_3 = 0 \end{array} \quad \begin{array}{c} \longleftrightarrow \\ \\ \end{array} \quad \begin{array}{c} 2\alpha_1 + 0.5\alpha_2 = 0 \\ -9.5\alpha_1 = 0 \\ \alpha_2 + 2\alpha_3 = 0 \end{array} \quad \begin{array}{c} \longleftrightarrow \\ \\ \end{array} \quad \begin{array}{c} 2(0) + 0.5\alpha_2 = 0 \\ \alpha_1 = 0 \\ \alpha_2 + 2\alpha_3 = 0 \end{array} \quad \begin{array}{c} \longleftrightarrow \\ \\ \end{array} \quad \begin{array}{c} \alpha_2 = 0 \\ \alpha_1 = 0 \\ 0 + 2\alpha_3 = 0 \end{array} \quad \begin{array}{c} \longleftrightarrow \\ \\ \end{array} \quad \begin{array}{c} \alpha_1 = 0 \\ \alpha_2 = 0 \\ \alpha_3 = 0 \end{array}$$

$R_2 - 0.5R_3$ 
 $R_2 - 5R_1$

The only linear combination is the trivial one. We can conclude that  $a_1, a_2, a_3$  are **linearly independent**. Thus, the three linearly independent vectors have a column rank of 3. Thus, we demonstrate that for matrix  $A$  the row rank is equal to the column rank. (This is unsurprising as for a matrix always the **row rank = column rank**)

- c) When applying the transpose to  $A$ , we are simply exchanging the rows for the columns. So checking for the column rank of  $A^T$  is the same as checking for the row rank of the original  $A$ . We have already done that and determined that the row rank of  $A$  is 3, so the column rank of  $A^T$  is 3. [...]



## Solution

# Exercise: Rank & Dimension

[...] Moreover, since the transpose only changes rows to columns and row and column rank are the same, even just knowing the original row or column rank would have been sufficient to know also the rank of the transposed matrix, which did not change compared to the original matrix.

- d) Yes, adding  $b_4$  will lead to the now four column vectors in  $B$  not being a basis for the 3-dimensional subspace anymore.
- (1) While the vectors remain all linearly independent and being able to reach any point in the 3-dimensional subspace, they now span a larger space – the four linearly independent vectors span  $\mathbb{R}^4$ . So, their span has changed. (If their span hadn't changed, the new vector would have been superfluous and, thus, it would also not be a basis anymore).
- (2) Moreover, we established that all basis for a specific vector space have the same size, so if the original basis consisted of 3 vectors, there cannot be a basis that has more or less vectors for the same vector space. Thus, having 4-vectors and being a basis for the same space is not possible.



# Some Key Take-aways (Lecture 13)

There can be different bases for the same vector space  $\mathcal{V}$  but all have the same size

If you have a basis, there is no smaller other basis

For a set of vectors  $S$ , if there is a linearly independent set of vectors  $B$  that span  $S$ , then  $B$  cannot be smaller than the set  $S$  [\[Morphing Lemma\]](#)

For a span of a set of vectors, we need at most this number of vectors

„**Dimension**“ is with respect to a vector space and „**rank**“ to a set of vectors

Every (finite) set of vectors has a subset that is a basis for the span of these vectors. For a set of  $D$ -vectors, the rank is at most  $|D|$  [\[Subset-Basis Lemma\]](#)

We can find a basis for the span

If we have some linearly independent vectors, then we can use them all (and some additional vectors) to create a basis [\[Superset-Basis Lemma\]](#)

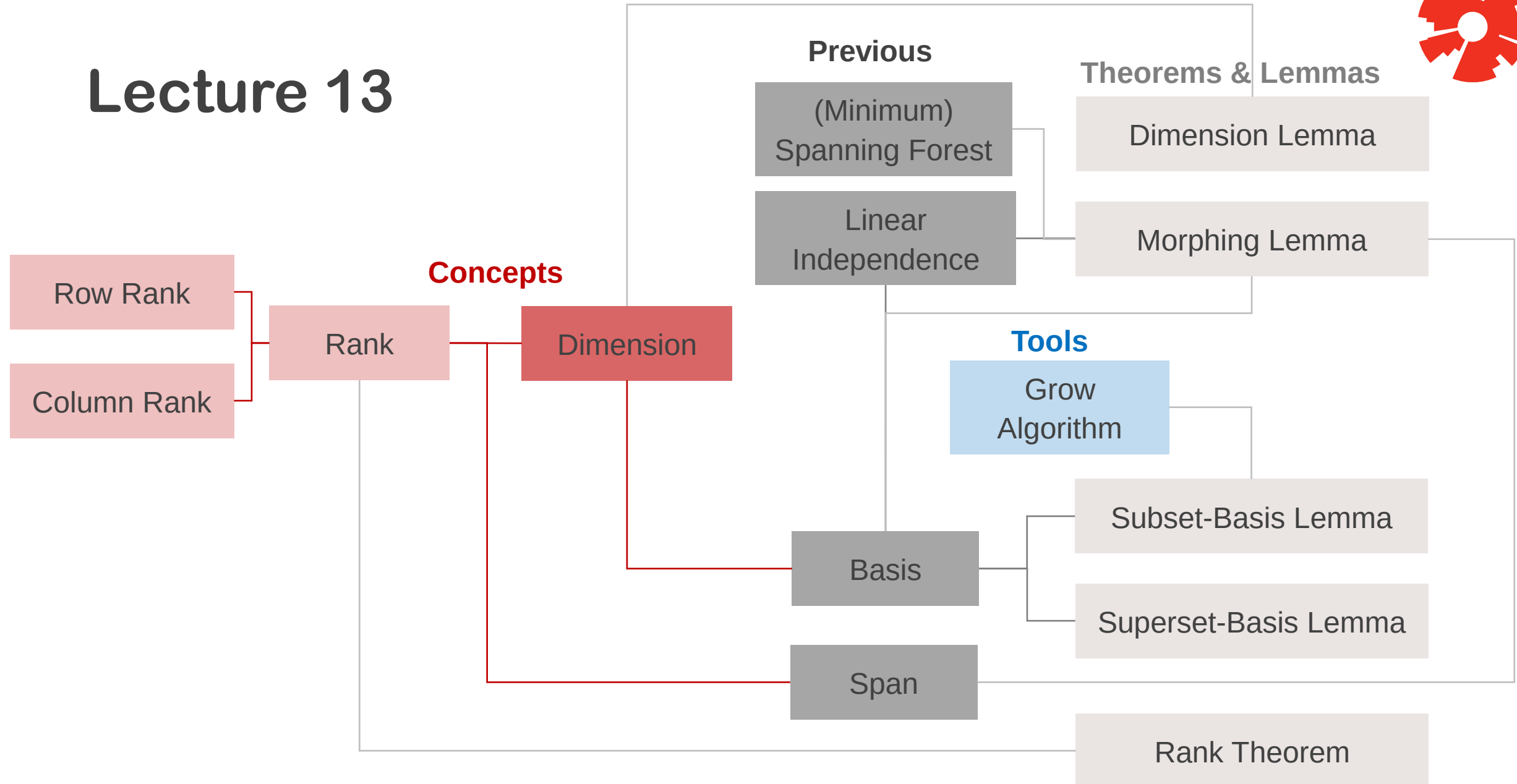
We can use such vectors for a basis

There are two ranks for a matrix (**row rank**, **column rank**) and they have the same value (same rank) [\[Rank Theorem\]](#)

We can focus on the column rank



# Lecture 13





# Thank You For Your Attention!



**Accomplishment unlocked:  
Lecture 13 Completed**



# APPENDIX



# Subset-Basis Lemma (II)

Revised version:

Initialize  $S = \emptyset$

Repeat while possible: select a vector  $\mathbf{v}$  in  $T$  that is not in  $\text{Span } S$ , and put it in  $S$ .



Differs from original:

- ▶ This algorithm stops when  $\text{Span } S$  contains every vector in  $T$ . (we can reach every vector in  $T$  with the vectors added to  $S$ )
- ▶ The original Grow algorithm stops only once  $\text{Span } S$  contains every vector in  $\mathcal{V}$ .

However, that's okay: when  $\text{Span } S$  contains all the vectors in  $T$ ,  $\text{Span } S$  also contains all linear combinations of vectors in  $T$ , so at this point  $\text{Span } S = \mathcal{V}$ .

Shows that original Grow algorithm can be guided to make same choices as this algorithm, so result is a basis.

QED



# Cardinality of a Vector Space over GF(2) (I)

Checksum function  $\mathbf{x} \mapsto [\mathbf{a}_1 \cdot \mathbf{x}, \dots, \mathbf{a}_{64} \cdot \mathbf{x}]$

Original “file”  $\mathbf{p}$ , transmission error  $\mathbf{e}$  so corrupted file is  $\mathbf{p} + \mathbf{e}$ .

Remember  
(some lectures ago)

What is probability that corrupted file has the same checksum as original?

If error is chosen according to uniform distribution,

$$\begin{aligned}
 & \text{Probability } (\mathbf{p} + \mathbf{e} \text{ has same checksum as } \mathbf{p}) \\
 &= \text{Probability } (\mathbf{e} \text{ is a solution to homogeneous linear system}) \\
 &= \frac{\text{number of solutions to homogeneous linear system}}{\text{number of } n\text{-vectors}} \\
 &= \frac{\text{number of solutions to homogeneous linear system}}{2^n}
 \end{aligned}$$

How to find number of solutions to a homogeneous linear system over  $GF(2)$ ?



# Cardinality of a Vector Space over $GF(2)$ (II)

How to find number of solutions to a homogeneous linear system over  $GF(2)$ ?

Solution set of a homogeneous linear system is a vector space. Question becomes

How to find out cardinality of a vector space  $\mathcal{V}$  over  $GF(2)$ ?

► Suppose basis for  $\mathcal{V}$  is  $\mathbf{b}_1, \dots, \mathbf{b}_n$ .

► Then  $\mathcal{V}$  is set of linear combinations

$$\beta_1 \mathbf{b}_1 + \dots + \beta_n \mathbf{b}_n$$

► Number of linear combinations is  $2^n$ . Since each of the  $n$  coefficients  $\beta_1, \dots, \beta_n$  is either 0 or 1

► By Unique-Representation Lemma, every linear combination gives a different vector of  $\mathcal{V}$ .

► Thus cardinality is  $2^{\dim \mathcal{V}}$ .



# Cardinality of a Vector Space over $GF(2)$ (III)

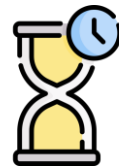
Cardinality of a vector space  $\mathcal{V}$  over  $GF(2)$  is  $2^{\dim \mathcal{V}}$ .

How to find dimension of solution set of a homogeneous linear system?

Write linear system as  $A\mathbf{x} = \mathbf{0}$ .

How to find dimension of the null space of  $A$ ?

Answers will come later.





# Simple Authentication revisited (I)

- Password is an  $n$ -vector  $\hat{\mathbf{x}}$  over  $GF(2)$
- **Challenge:** Computer sends random  $n$ -vector  $\mathbf{a}$
- **Response:** Human sends back  $\mathbf{a} \cdot \hat{\mathbf{x}}$ . Repeated until Computer is convinced that Human knows password  $\hat{\mathbf{x}}$ .

Eve eavesdrops on communication, learns  $m$  pairs

$\mathbf{a}_1, b_1$  (query, response)

$\vdots$

$\mathbf{a}_m, b_m$

such that  $b_i$  is right response to challenge  $\mathbf{a}_i$

Then Eve can calculate right response to any challenge in  $\text{Span}\{\mathbf{a}_1, \dots, \mathbf{a}_m\}$ :

Suppose  $\mathbf{a} = \alpha_1 \mathbf{a}_1 + \dots + \alpha_m \mathbf{a}_m$  Then right response is  $\alpha_1 b_1 + \dots + \alpha_m b_m$

My first answer is  $[0, 1, 0, 1, 0, 0]$   
and my second answer is ...



I know all answers he gave and all  
answers that I can infer from that!





# Simple Authentication revisited (II)

## Queries

**Fact:** Probably rank  $[\mathbf{a}_1, \dots, \mathbf{a}_m]$  is not much less than  $\min\{m, n\}$ . **No. of Queries**, **Password Length**  
 $n \times m$

Once  $m > n$ , probably Span  $\{\mathbf{a}_1, \dots, \mathbf{a}_m\}$  is all of  $GF(2)^n$  so Eve can respond to **any** challenge.  
 (→ many queries)

**Also:** The password  $\hat{\mathbf{x}}$  is a solution to

$$\underbrace{\begin{bmatrix} \mathbf{a}_1 \\ \vdots \\ \mathbf{a}_m \end{bmatrix}}_A \begin{bmatrix} \mathbf{x} \end{bmatrix} = \underbrace{\begin{bmatrix} b_1 \\ \vdots \\ b_m \end{bmatrix}}_{\mathbf{b}}$$

Solution set of  $A\mathbf{x} = \mathbf{b}$  is  $\hat{\mathbf{x}} + \text{Null } A$

With enough queries, other solutions (*Null A*) drop away, leaving only one solution ( $\hat{\mathbf{x}}$ )!

Once rank  $A$  reaches  $n$ , columns of  $A$  are linearly independent so Null  $A$  is trivial, so only solution is the password  $\hat{\mathbf{x}}$ , so **Eve can compute the password** using solver.

